BSAN 750 Data Mining and Machine Learning

Lecture 3. Linear Regression[1]

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[1] Partially based on Hastie, et al. (2009) ESL, and James, et al. (2013) ISLR

Linear Regression – a fundamental learning algorithm

- Supervised learning method
- It assumes the dependence of Y on X is linear
- Largely used in many disciplines
- Simple and interpretable
- Fundamental in data science

Linear Regression Models

- Simple linear regression

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- Multiple linear regression

$$Y = \beta_0 + \beta_1 X + \ldots + \beta_p X_p + \epsilon$$

- Y: dependent variable (response, outcome)
- X's: independent variable (covariates, explanatory variable)
- β's: regression coefficients
- ϵ: random error (irreducible error)

Linear Regression Models

- Using matrix format

$$Y = \mathbf{X}\boldsymbol{\beta} + \epsilon$$

- $\mathbf{X}$ is called design matrix with first column being 1's
- The *estimated linear regression model* is

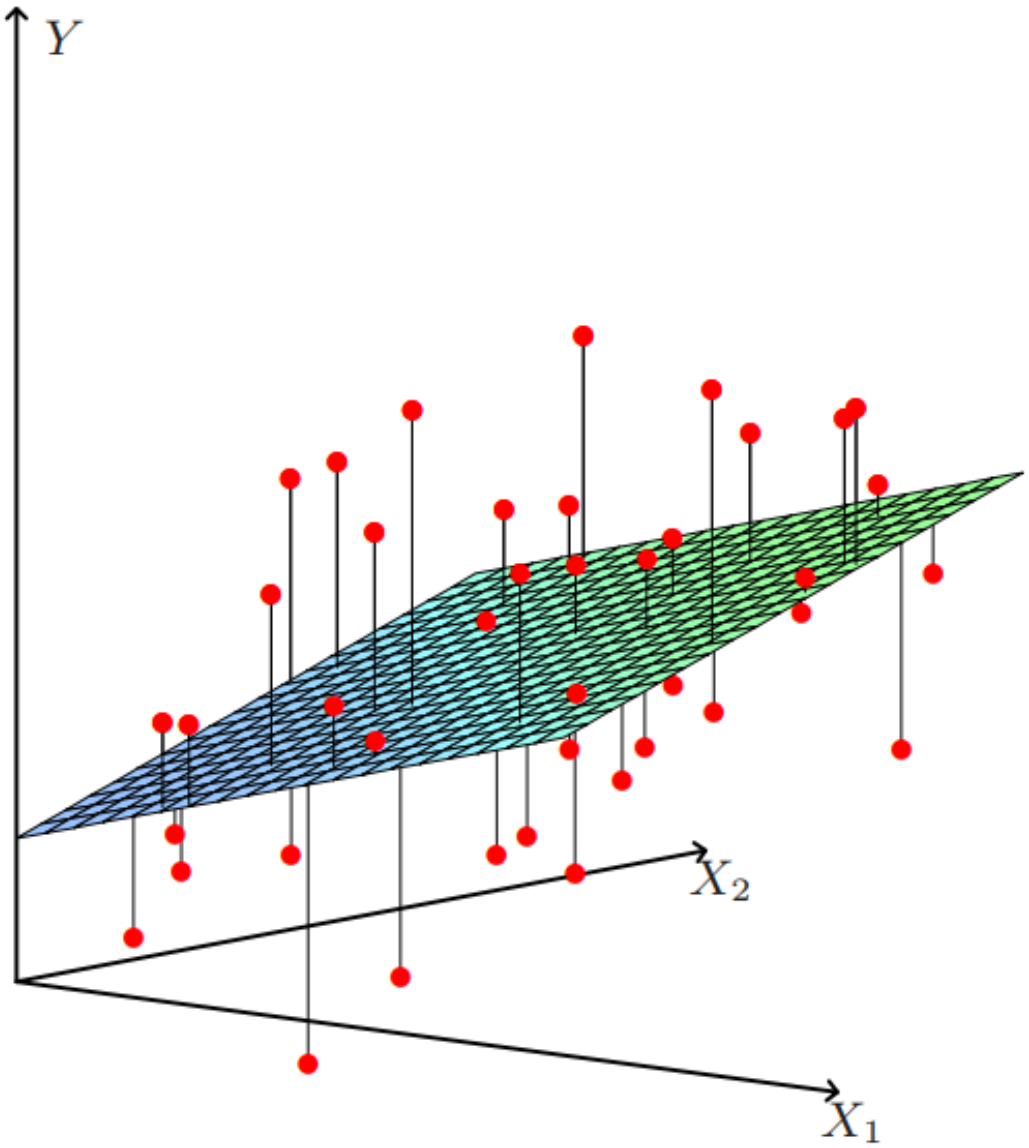
$$\hat{Y} = \mathbb{E}(Y|X) = \mathbf{X}\hat{\boldsymbol{\beta}}$$

- Goal: estimate regression coefficient β

Model Assumptions

- $\mathbb{E}(Y|X)$ is a linear function of X or its basis expansion such as X_1^2, $X_2^3, \ldots$
- The error term $\{\epsilon_i, \ldots, \epsilon_n\} \overset{i.i.d.}{\sim} N(0, \sigma^2)$

Least Square Solution



Least Square Solution

- We want to minimize residual sum squares

$$
\begin{aligned}
RSS(\beta) &= \sum_{i=1}^{n} (y_i - \mathbf{x}_i^T \beta)^2 \\
&= (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta)
\end{aligned}
$$

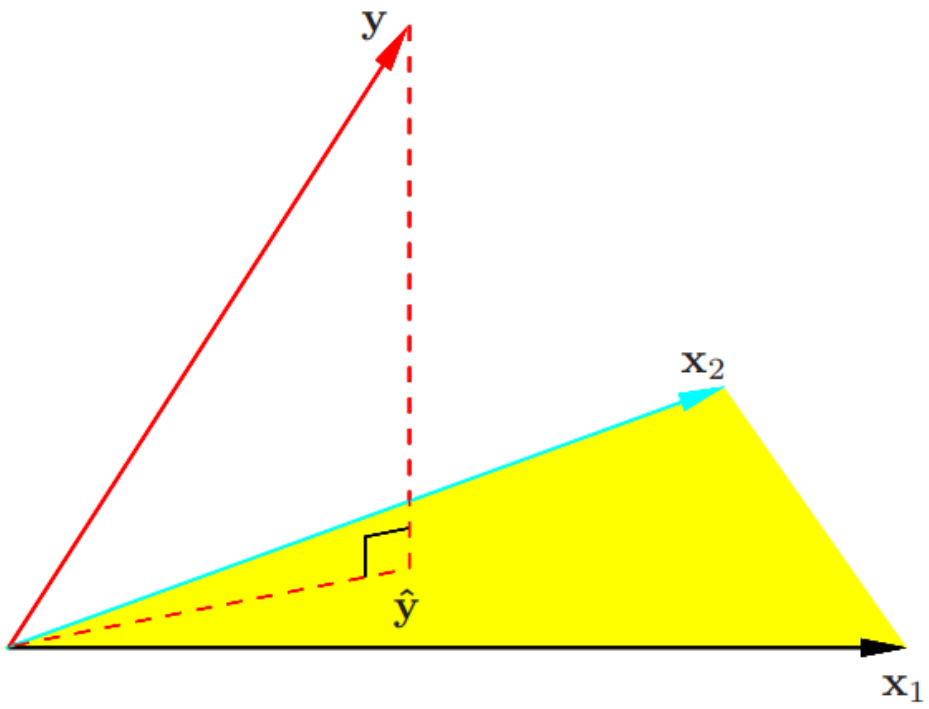
- Take first-order derivative with respect to β and set to 0

$$
\begin{aligned}
0 &= \frac{\partial RSS(\beta)}{\partial \beta} = -2\mathbf{X}^T(\mathbf{y} - \mathbf{X}\beta) \\
\mathbf{X}^T\mathbf{y} &= \mathbf{X}^T\mathbf{X}\beta
\end{aligned}
$$

- This is called *normal equation*.

Least Square Solution

- By assuming $p < n$, the solution is $\hat{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$
- The predicted value is $\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$
- $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is called hat matrix or projection matrix



Model Assessment – R Square and MSE

- It is proportion of variation in Y explained by the model

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

  R^2 increases monotonically as number of explanatory variable increasing.

- Adjusted R^2

$$R^2_{adj} = 1 - \frac{n-1}{n-p-1}\frac{RSS}{TSS}$$

- Mean Squared Error (MSE)

$$MSE = \frac{1}{n-p-1} \times RSS$$

  It is an unbiased estimate of σ^2 for irreducible error ϵ.

Model Assessment – AIC, BIC, Cp, CV

- Akaike information criterion (AIC), the smaller the better

$$AIC = -2\log(\hat{L}) + 2p$$

- Bayesian information criteria (BIC), the smaller the better

$$BIC = -2\log(\hat{L}) + \log(n)p$$

  where $\hat{L}$ is estimated likelihood function
- Mellow's C_p is the same as AIC for linear regression
- Cross-validation error (CV score)

Hypothesis Testing – Test individual coefficients

Is a specific X relevant?

- Testing for individual β
  - H_0: $\beta_j = 0$; H_1: $\beta_j \neq 0$
  - Using T-test since the true variance is unknown

$$T = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)} = \frac{\hat{\beta}_j}{\hat{\sigma}\sqrt{v_j}} \sim t_{n-p-1}$$

  where v_j is the jth diagonal element of $(\mathbf{X}^T\mathbf{X})^{-1}$
  - Reject H_0 if p-value $< \alpha$ or $|T| > T_{1-\alpha}^{(n-p-1)}$
- Confidence interval: $\hat{\beta} \pm se(\hat{\beta}) \times T_{1-\alpha}^{(n-p-1)}$

Hypothesis Testing – Test multiple coefficients

- F-test for overall significance
  - H_0: $\beta_1 = \ldots = \beta_p = 0$; H_1: at least one $\beta \neq 0$
  - F statistics

$$F^* = \frac{(TSS - RSS)/p}{RSS/(n-p-1)} \sim F_{p,n-p-1}$$

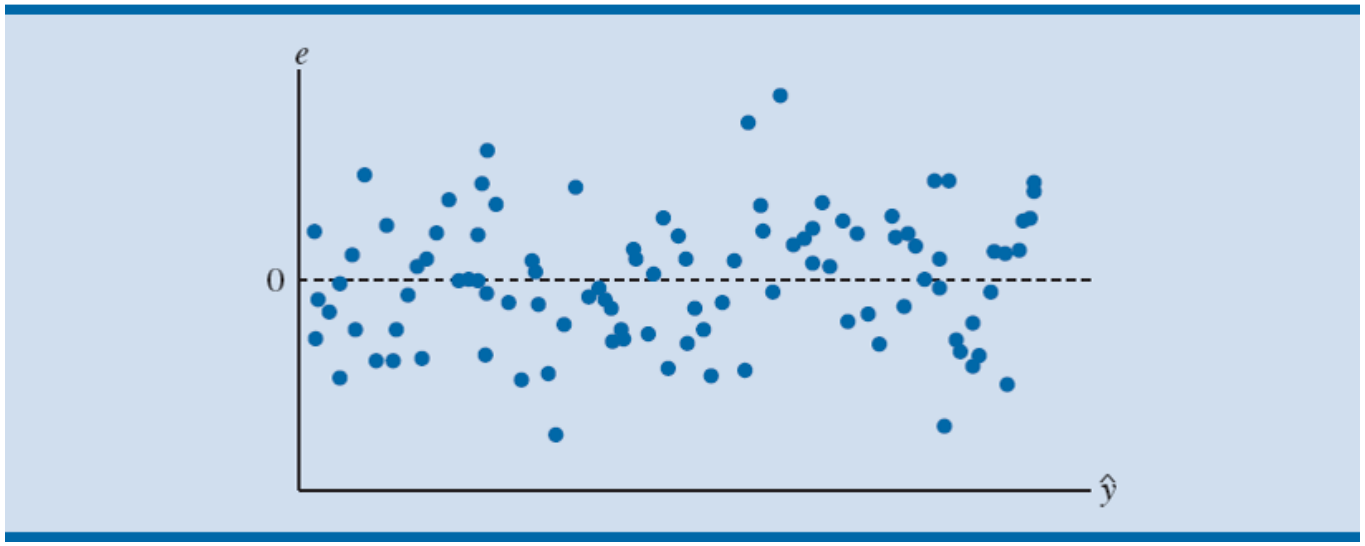
where $TSS = \sum_{i=1}^{n}(y_i - \bar{y})^2$, is total sum squares

Bootstrap

- Resampling method
- A powerful tool to quantify uncertainty
  - standard error
  - confidence interval
- Random sampling with replacement
- Example:
  - train a model with 1000 bootstrap samples
  - store all the parameter estimates
  - calculate standard error and confidence interval

Model Diagnostics

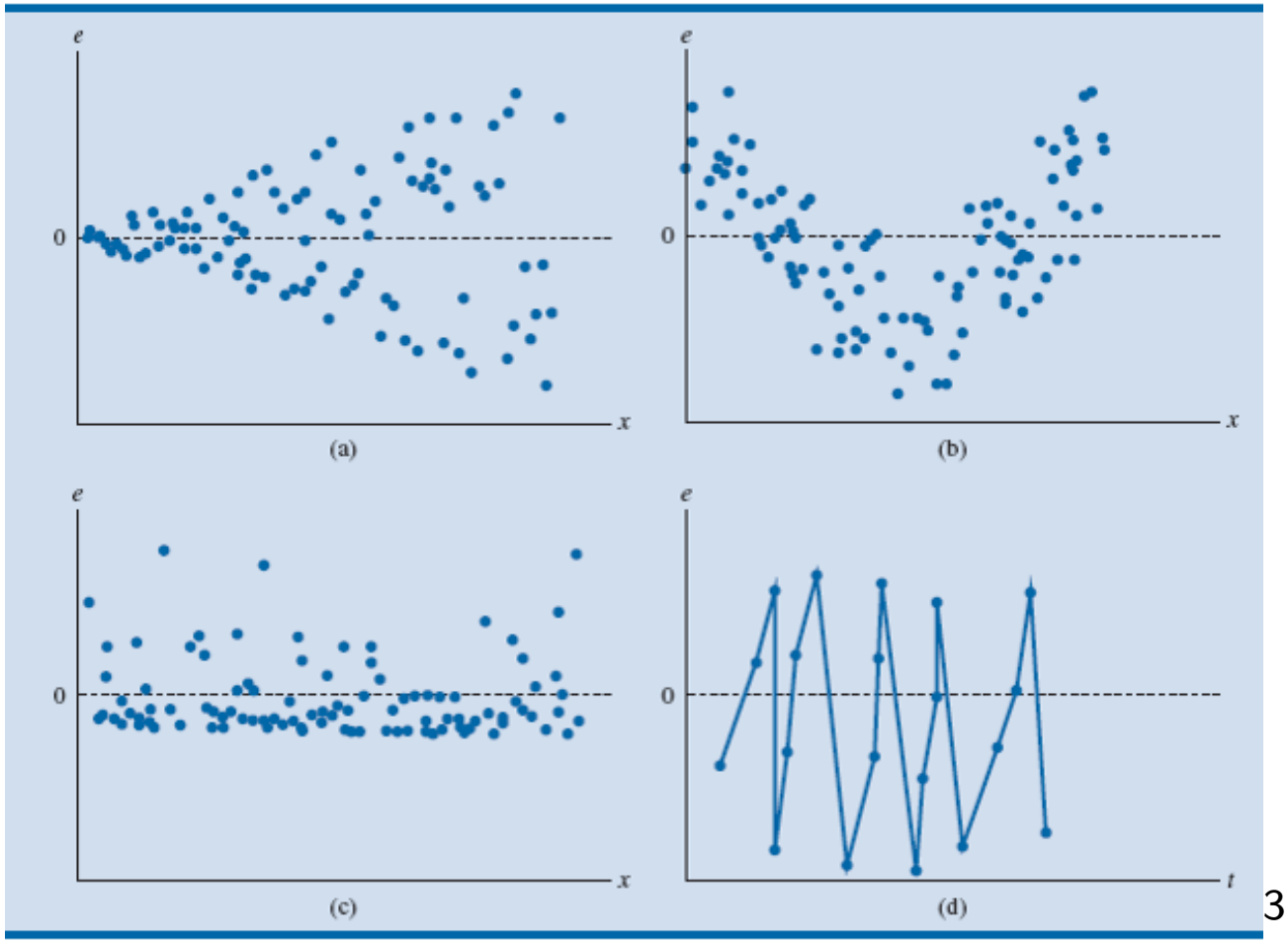
- Check the assumptions on the error term
  - Independent normal distribution?
  - $\mathbb{E}(\epsilon_i) = 0$?
  - $Var(\epsilon_i) = \sigma^2$ =constant?
- Residual plot (an ideal residual plot looks like this)[2]



[2]source: Camm, et al., *Essentials of Business Analytics*

Residual plot

Which type of assumption is violated?



[3]

[3] source: Camm, et al., *Essentials of Business Analytics*

Other Diagnostic Plots

- Normal Quantile-Quantile Plot
  - It plots the standardized residual vs. theoretical quantiles
  - An easy way to visually test the normality assumption
  - If residual follows normal distribution, you should expect all dots lie on the diagonal straight line.
- Residual-Leverage Plot
  - This plot checks if there are any influential points, which could alter your analysis by excluding them
  - The points that lie outside the dashed line, Cook's distance, are considered as influential points